

MIP_CODING_STANDARDS.md

Mainframe Intelligence Platform

Coding Standards

Version: 1.0

Purpose

Define coding standards for all MIP contributors.

Goals:

- Consistency
 - Maintainability
 - Testability
 - Scalability
 - Readability
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Guiding Principles

Code is read more often than written.

Prefer:

- Clarity
- Simplicity
- Explicitness

Avoid:

- Clever code
 - Hidden logic
 - Premature optimization
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Python Standards

Version:

Python 3.13+

Naming Standards

Files

snake_case.py

Example:

program_parser.py

Classes

PascalCase

Example:

ProgramParser

Functions

snake_case

Example:

extract_program_id()

Constants

UPPER_CASE

Example:

MAX_FILE_SIZE

Type Hints

Mandatory.

Example:

```
def parse_program(path: str) -> Program:
    pass
```

Pydantic

All metadata models must use Pydantic.

Never use raw dictionaries as domain entities.

Logging

Use structured logging.

Never use print().

Exceptions

Raise meaningful exceptions.

Bad:

```
raise Exception()
```

Good:

```
raise MetadataValidationError()
```

Documentation

Every public class and function requires docstrings.

Complexity Rules

Function:

≤ 50 lines preferred

Class:

Single responsibility

Testing Requirements

Every production module:

- Unit tests
- Edge cases
- Failure cases

Coverage:

Minimum 80%

Target 90%

Review Checklist

- Naming compliant?
- Typed?
- Tested?
- Documented?
- Logged?
- Maintainable?